

BundleWise: A Retail Analytics Tool for Smart Bundling and Profit-Driven Pricing

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A Project report submitted in partial fulfillment of the requirements for the award of the degree of Master of Science (Data Science) of CHRIST (Deemed to be University)

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CERTIFICATE

*This is to certify that the report titled **BundleWise: A Retail Analytics Tool for Smart Bundling and Profit-Driven Pricing** is a bonafide record of work done by **Mansi Sapariya (2448036)** of **CHRIST (Deemed to be University)**, Bengaluru, in partial fulfillment of the requirements of IV Trimester MSc (Data Science) during the academic year 2025-26.*

Head of the Department

Project Guide

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ABSTRACT

BundleWise is a retail analytics system developed to empower small and medium-sized businesses with intelligent product bundling, dynamic pricing, and actionable insights from transactional data. While retailers generate large volumes of sales data every day, much of it remains underutilized, and existing solutions typically focus on either bundling or pricing while being designed for large enterprises, leaving smaller retailers without affordable and self-service alternatives. This project addresses that gap by integrating three core modules: bundle recommendation to identify frequently purchased product combinations, pricing optimization to suggest profit-driven strategies for bundles and individual products, and an interactive analytics dashboard to deliver insights into product performance, customer behavior, and advanced sales trends. The system accepts data in CSV, XLS, or PDF formats, validates and preprocesses inputs automatically, and enables results to be exported in user-friendly PDF or Excel reports. Developed using Python, Flask, Pandas, and Plotly, the platform emphasizes usability through an intuitive dashboard with tooltips, filters, and advanced charts that make analytics accessible to non-technical users. The outcome is a functional prototype capable of generating bundle recommendations, optimized pricing strategies, and actionable insights to support data-driven decision-making. Future enhancements include integration with point-of-sale (POS) systems, incorporation of forecasting models for demand prediction, and the development of real-time analytics for faster retail intelligence. By bridging academic methodologies with practical retail needs, BundleWise delivers a scalable, cost-effective, and India-first solution that supports modern retailers in improving profitability, enhancing customer engagement, and gaining a competitive edge through data-driven decision-making.

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ABBREVIATIONS

Abbreviation	Full Form
MBA	Market Basket Analysis
SPM	Sequential Pattern Mining
KPI	Key Performance Indicator
PDF	Portable Document Format
XLS	Excel Spreadsheet
SDLC	Software Development Life Cycle
CRISP-DM	Cross-Industry Standard Process for Data Mining
RM	Research Methodology
SOP	Standard Operating Procedures
SDD	System Design Document
SRS	Software Requirements Specification
SKU	Stock Keeping Unit
CSV	Comma-Separated Values
UI	User Interface
POS	Point of Sale
ERP	Enterprise Resource Planning

API	Application Programming Interface
PII	Personally Identifiable Information
GDPR	General Data Protection Regulation
RBAC	Role-Based Access Control
KDD	Knowledge Discovery in Databases
SEMMA	Sample, Explore, Modify, Model, Assess

CHAPTER 1: INTRODUCTION

This chapter introduces the **BundleWise project**, its motivation, objectives, and overall problem domain. It explains why bundling and pricing optimization are critical for retailers, outlines the scope and significance of the system, and states the project's aims. The chapter also provides a brief overview of the methodology followed and the expected outcomes of the project.

1.1 PROBLEM DESCRIPTION

Retailers generate large volumes of transaction data every day. However, much of this data is not fully used to support decision-making. Two key business areas that benefit from data-driven insights are product bundling and pricing optimization.

- Bundling helps businesses identify products that are often purchased together and combine them into attractive offers.
- Pricing optimization ensures products and bundles are priced in a way that balances profitability with customer appeal.

In addition, retailers also need general analytics to better understand their product performance, customer behavior, and overall sales trends. Without automated systems, retailers face:

- Missed cross-selling and upselling opportunities.
Prices that are either too high (reducing sales) or too low (reducing profit).
- Limited understanding of customer preferences and product contribution.
- Dependence on manual reporting which is slow and error-prone.

This creates the need for a single platform that can not only recommend bundles and optimize prices but also provide a comprehensive analytics dashboard to visualize sales data in a clear and useful manner.

1.2 EXISTING SYSTEMS

Several ERP and SaaS-based analytics platforms exist in the market, but most are designed for large organizations. They are either **bundling-only** or **pricing-only**, often lacking a self-service platform that is affordable and user-friendly. Furthermore, most of these systems are either consulting-based or too complex for smaller retailers.

Table 1.1 - Comparison of Existing Systems with BundleWise

Company / Tool	What It Does	Focus Area	How It Differs from BundleWise
Increasingly (UK)	Provides AI-powered product bundling to increase basket size (Average Order Value) for e-commerce retailers.	Online and e-commerce platforms.	Specializes in bundling only. Does not offer price optimization or advanced analytics. BundleWise adds both pricing and interactive dashboards.
Revionics (US)	Offers lifecycle pricing, including base price, promotions, and markdown optimization.	Large-scale enterprise retailers.	Strong in pricing optimization , but has no features for bundling or transaction-level insights. BundleWise merges both bundling and pricing capabilities.

Competera (Global)	An AI-driven dynamic pricing platform that adjusts prices based on demand and competition.	Global retail and omnichannel brands.	Entirely pricing-focused . Does not analyze product combinations or customer purchase patterns. BundleWise provides both MBA/SPM bundling and price optimization .
Pricefx (Global)	A cloud-based SaaS platform for price management and optimization.	Large B2B and retail enterprises	A complex enterprise tool that is pricing-only . BundleWise is simpler, cost-effective, and India-first , while also including bundling insights.
SPEC INDIA (India)	Delivers custom Market Basket Analysis (MBA) and dashboards through consulting projects using BI tools.	Retail analytics consulting.	Provides MBA as a service , not as a self-service product. BundleWise is designed as a ready-to-use platform that retailers can operate independently.
Cornerstone Solutions (Global)	AI-powered MBA service with POS integration for retailers.	Retail service providers.	Provides bundling analytics, but as a consulting solution . BundleWise is a self-serve platform with built-in dashboards.

From the above comparison, it is clear that while many tools specialize either in **bundling** or **pricing**, *BundleWise* is unique in combining **bundling, pricing optimization, and analytics dashboards** into one platform. Additionally, *BundleWise* is designed to be **affordable, self-service, and particularly suited for small and medium businesses**, filling a gap that current solutions leave unaddressed.

1.3 PROJECT SCOPE

The project *BundleWise* focuses on three modules: bundling, price optimization, and analytics dashboards.

1.3.1 INCLUDED IN SCOPE:

- **Data Upload and Preprocessing:** Upload transaction data in CSV/XLS/PDF format, validate required fields, and clean inconsistencies.
- **Market Basket Analysis (MBA):** Identify products that are frequently purchased together within transactions.
- **Sequential Pattern Mining (SPM):** Find product purchase sequences across transactions over time.
- **Pricing Optimization:** Suggest pricing strategies for bundles and individual products to maximize revenue.
- **Analytics Dashboard:** Provide interactive visualizations across multiple subtabs:
 - **Summary:** High-level overview of revenue, number of products, total transactions, and average order value.
 - **Product Performance:** Insights into sales distribution, top-selling products, and product contribution to revenue.
 - **Customer Insights:** Analysis of customer purchasing patterns, order frequency, and loyalty behavior.
 - **Advanced Analytics:** Deeper visualizations to explore correlations, sales trends, and transaction patterns.
- **Report Generation:** Export analysis and insights in PDF or Excel formats.

1.3.2 EXCLUDED FROM SCOPE:

- Forecasting of sales or demand.
- Real-time POS/ERP integration (the system is designed for CSV uploads only).

1.4 OBJECTIVES

The objectives of *BundleWise* are to:

- Build a bundling module that discovers frequent itemsets and patterns using MBA and SPM.
- Implement a pricing optimization module to recommend profit-driven prices.
- Develop an analytics dashboard with subtabs (Summary, Product, Customer, Advanced) to provide comprehensive sales insights.
- Create an intuitive interface with tooltips and charts so non-technical users can easily interpret results.
- Keep the design modular and scalable to add new analytics features in the future.

1.5 MOTIVATION

Modern retail is highly competitive, and businesses must make quick and informed decisions to succeed. Large organizations have access to advanced analytics, but smaller businesses lack simple and affordable options.

The motivation for *BundleWise* comes from the need to:

- Simplify data analytics by offering clear dashboards with product, customer, and advanced insights.
- Directly improve retail outcomes through smart bundling and optimized pricing.
- Bridge academic techniques with practical value by applying methods like MBA and SPM to real transaction data.
- While globally relevant, *BundleWise* is particularly unique in India, where no integrated tool currently exists that combines bundling, pricing optimization, and interactive dashboards.

CHAPTER 2: SYSTEM ANALYSIS

This chapter reviews the existing tools, systems, and research studies related to retail analytics, market basket analysis, sequential pattern mining, and dynamic pricing. It highlights the strengths and gaps of current solutions, compares them with the proposed system, and positions BundleWise as a more comprehensive, lightweight, and retailer-focused alternative.

2.1 FUNCTIONAL SPECIFICATION

The functional specifications of *BundleWise* describe what the system is expected to do and how each component contributes to the overall goal. Based on the SRS and design documents, the system has the following core functions:

2.1.1 DATA UPLOAD AND PREPROCESSING

- Retailers upload sales data in CSV/XLS/PDF format.
- The system validates key fields such as Transaction ID, Product, Quantity, and Price.
- Missing or invalid data entries are cleaned or flagged.
- Preprocessed data is stored temporarily for analysis.

2.1.2 BUNDLING MODULE

- Uses **Market Basket Analysis (MBA)** to find frequent itemsets from transaction data.
- Applies **Sequential Pattern Mining (SPM)** to detect purchasing sequences over time.
- Generates bundle recommendations based on product association strength and frequency.

2.1.3 PRICING OPTIMIZATION MODULE

- Takes identified bundles as input.
- Suggests pricing strategies to maximize revenue or move slow-selling inventory.

- Ensures flexibility so retailers can adapt recommendations based on business priorities.

2.1.4 ANALYTICS DASHBOARD

The dashboard provides an interactive visualization layer with the following subtabs:

- **Summary:** Displays KPIs such as total revenue, total transactions, number of products, and average order value.
- **Product Performance:** Shows top-performing products, contribution to revenue, and sales distribution.
- **Customer Insights:** Provides details on purchasing behavior, frequency of transactions, and customer loyalty.
- **Advanced Analytics:** Offers deeper trend visualizations such as monthly sales trends, correlations, and breakdowns.

Tooltips and hover-based info panels make the insights more user-friendly.

2.1.5 REPORTING

- Analysis results can be exported in **PDF** and **Excel** formats.
- Exports include charts, tables, and recommendations, making them suitable for business reporting.

2.2 BLOCK DIAGRAM

The high-level block diagram of the system can be represented as follows:

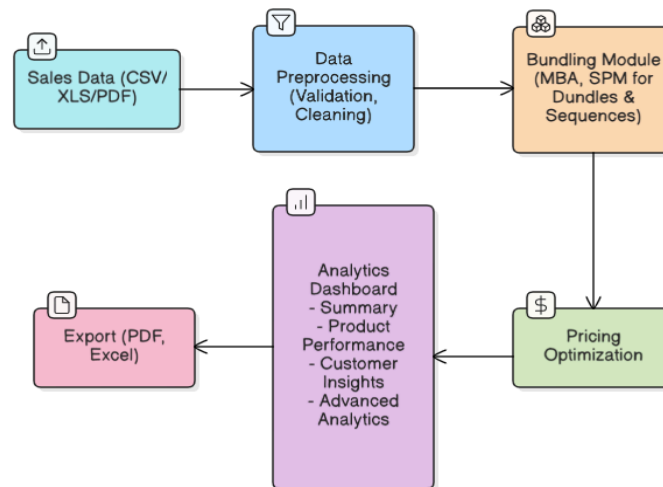


Fig 2.1 - Block Diagram

2.3 SYSTEM REQUIREMENTS

The system requirements are divided into **functional** and **non-functional** specifications.

2.3.1 FUNCTIONAL REQUIREMENTS

- The system must allow CSV uploads of sales data.
- It must validate and clean uploaded data.
- It must generate bundle recommendations using MBA and SPM.
- It must suggest optimized pricing for bundles.
- It must provide an analytics dashboard with Summary, Product, Customer, and Advanced views.
- It must allow exporting reports in PDF and Excel formats.

2.3.2 NON-FUNCTIONAL REQUIREMENTS

- **Usability:** The dashboard must be intuitive and accessible for non-technical users.
- **Performance:** The system should handle moderately large datasets without

significant delays.

- **Scalability:** Architecture should support the addition of new analytics features in the future.
- **Portability:** The solution should be deployable on a web server or local machine with minimal setup.
- **Reliability:** Data processing should maintain integrity, with proper error handling for incorrect file formats.

CHAPTER 3: SYSTEM DESIGN

System design is one of the most critical phases of software development as it lays down the blueprint for how the system will function and how users will interact with it. In the case of *BundleWise*, system design ensures that uploaded sales data is processed efficiently, bundling and pricing recommendations are generated accurately, and insights are delivered through an interactive analytics dashboard.

This chapter describes the architecture, modules, data flow, entity relationships, interface designs, and reporting framework implemented in *BundleWise*.

3.1 SYSTEM ARCHITECTURE

The architecture of *BundleWise* is modular, lightweight, and designed to ensure smooth integration of multiple components. It allows retailers to upload sales data in different formats (CSV, Excel, or PDF), process this data into a structured format, perform bundling and pricing optimization, and finally present the results on an interactive analytics dashboard.

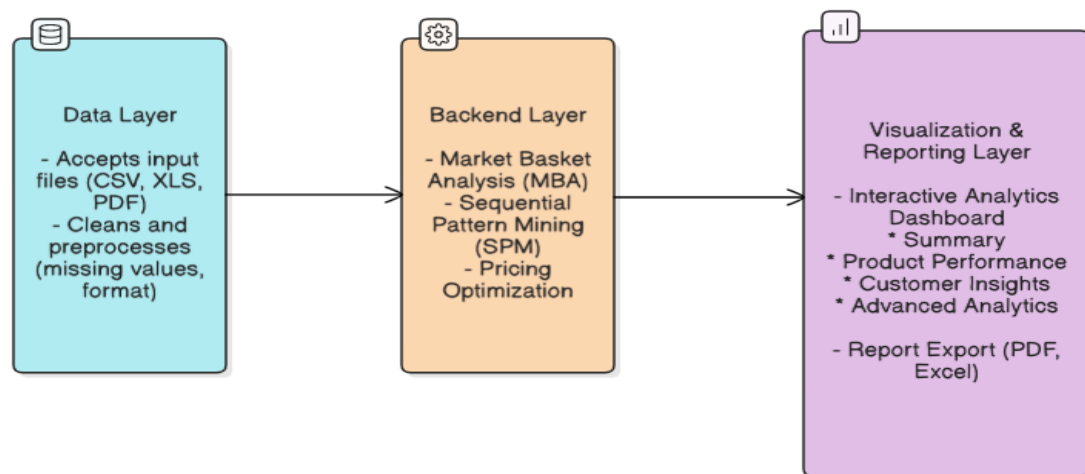


Fig 3.1 - System Architecture Diagram

The architecture is divided into three main layers:

3.1.1 DATA LAYER

- Accepts input data in CSV, XLS, or PDF format.
- Parses and validates uploaded files.
- Preprocesses data (handling missing values, converting formats).

3.1.2 BACKEND LAYER

- Performs **Market Basket Analysis (MBA)** and **Sequential Pattern Mining (SPM)** for bundle creation.
- Executes **Pricing Optimization** to recommend competitive and profitable bundle prices.

3.1.3 VISUALIZATION AND REPORTING LAYER

- Provides an **analytics dashboard** with multiple subtabs: Summary, Product Performance, Customer Insights, and Advanced Analytics.
- Generates reports in **PDF and Excel** formats for download.

3.2 MODULE DESIGN

BundleWise has been developed using a modular design, where each module is responsible for a well-defined function.

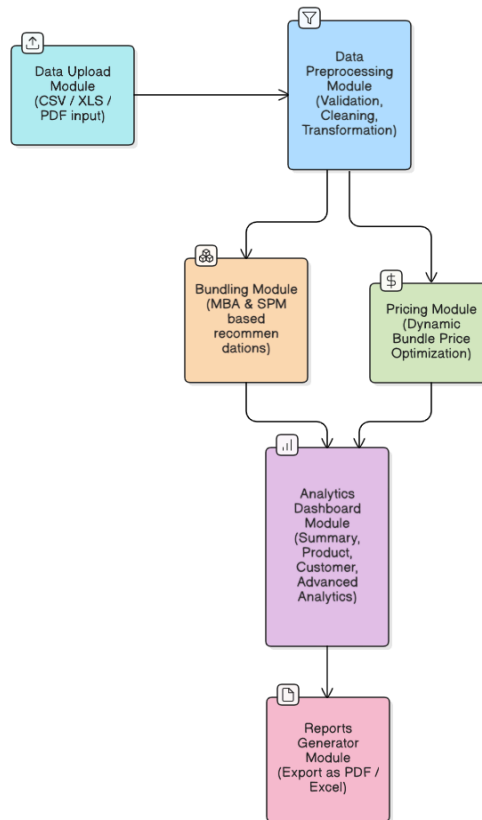


Fig 3.2- Module Interaction Diagram

3.2.1 DATA UPLOAD & PREPROCESSING MODULE

- Allows users to upload files in CSV, Excel, or PDF format.
- Parses and validates input files to check for missing or invalid values.
- Cleans and structures the data into Pandas DataFrames for further analysis.

3.2.2 BUNDLING MODULE

- Uses Market Basket Analysis (MBA) to identify frequently purchased product combinations.

- Applies Sequential Pattern Mining (SPM) to detect order-based purchase patterns.
- Produces bundles with metrics like support, confidence, and lift.

3.2.3 PRICING OPTIMIZATION MODULE

- Suggests competitive and optimized prices for bundles.
- Considers revenue maximization while ensuring affordability for customers.

3.2.4 ANALYTICS DASHBOARD MODULE

- Offers interactive insights through four subtabs:
 - **Summary** → Key metrics like total revenue, transactions, and average order value.
 - **Product Performance** → Product-level contribution to revenue, most/least sold products.
 - **Customer Insights** → Customer behavior, loyalty, and repeat purchase analysis.
 - **Advanced Analytics** → Trends, correlations, seasonal/monthly breakdowns.

3.2.5 REPORTING MODULE

- Generates reports in **PDF and Excel** formats.
- Summarizes bundles, pricing suggestions, and analytics insights.

3.3 DATA ORGANIZATION

BundleWise does not use a traditional database management system. Instead, it works with uploaded files and processes them into **temporary in-memory structures** using Pandas. This makes the system lightweight and easy to deploy while still ensuring structured data organization.

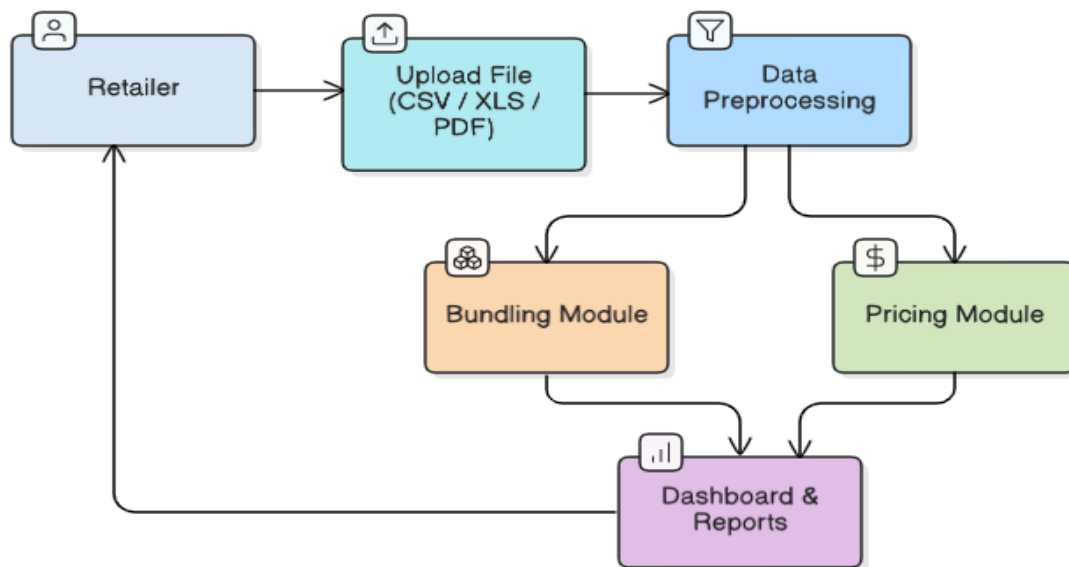


Fig 3.3 - Data Flow Diagram

3.4 INTERFACE DESIGN

The interface design of *BundleWise* emphasizes **clarity, usability, and accessibility** for retailers. Since the primary users may not have a strong technical background, the interface has been developed to be intuitive while providing powerful data-driven insights.

The interface is organized into four major sections:

1. **Data Upload & Validation**
2. **Bundle Recommendations & Pricing Optimization**
3. **Analytics Dashboard** (with subtabs for summary, product performance, customer insights, and advanced analytics)
4. **Reports Export**

3.4.1 DATA UPLOAD & VALIDATION SCREEN

- Users can upload their sales data in multiple formats such as **CSV, XLS, or PDF**.
- The system automatically validates the input for missing values, incorrect fields, and formatting inconsistencies.
- If the uploaded file passes validation, the data is preprocessed and made ready for further analysis.
- Error messages are displayed for invalid uploads, ensuring data quality before analysis begins.

3.4.2 BUNDLE RECOMMENDATIONS & PRICING OPTIMIZATION SCREEN

The recommendation screen highlights **bundles of products** that customers frequently buy together, along with optimized pricing strategies.

3.4.2.1 COMPLEMENTARY COLLECTION OFFERS

- Generated using Market Basket Analysis (MBA).
- Identifies combinations of items that are often purchased in the same transaction.

- Provides retailers with opportunities to promote these items together.
- Example: *“If customers often buy tea and sugar together, they can be offered as a discounted bundle.”*

3.4.2.2 BUY NOW, SAVE NEXT OFFERS

- Generated using Sequential Pattern Mining (SPM).
- Identifies products that are often purchased in sequence over multiple visits.
- Encourages repeat sales through targeted cross-sell offers.
- Example: *“A customer who buys a mobile phone may later purchase accessories, which can be promoted in advance.”*

3.4.2.3 DISPLAYED ON EACH RECOMMENDATION CARD:

- Suggested bundle price compared with the sum of individual prices.
- Potential customer savings.
- Validation metrics such as confidence, support, and lift.
- Expected impact on revenue.

This ensures recommendations are **practical, transparent, and beneficial** for both customers and retailers.

3.4.3 ANALYTICS DASHBOARD

The **Analytics Dashboard** serves as the **central hub of insights** in *BundleWise*. It transforms processed sales data into meaningful **visualizations, KPIs, and trends**, allowing retailers to explore business performance from multiple angles.

The dashboard is structured into **four subtabs**:

3.4.3.1 SUMMARY TAB

- Provides a high-level overview of the business.
- Displays KPIs such as **total revenue, number of products, transaction counts, and average order values**.

- Visualizations include **revenue contribution by category**, **sales trend charts**, and **top-performing products**.

3.4.3.2 PRODUCT PERFORMANCE TAB

- Focuses on analyzing product-level performance.
- Highlights best-selling products, revenue concentration, price ranges, and average selling prices.
- Visualizations include **price vs volume performance charts**, **category-wise revenue breakdowns**, and **distribution plots**.
- Example: *A retailer can identify that a few products contribute to most of the revenue, helping refine inventory strategy.*

3.4.3.3 CUSTOMER INSIGHTS TAB

- Provides insights into **customer behavior and loyalty**.
- Includes KPIs such as total customers, average customer value, and repeat purchase rates.
- Visualizations include **spending distribution charts**, **transaction frequency patterns**, **customer segmentation analysis**, and **new vs repeat purchase behavior**.
- Example: *Retailers can identify high-value customers and target them with loyalty offers.*

3.4.3.4 ADVANCED ANALYTICS TAB

- Enables deeper analysis into **cross-selling**, **seasonal patterns**, and **profitability**.
- Includes KPIs such as cross-sell opportunities, average items per transaction, and seasonal variation in sales.
- Visualizations include **cross-sell heatmaps**, **seasonal trend graphs**, and **profitability breakdowns**.
- Example: *The dashboard may reveal that sales peak during festive months, guiding retailers to plan promotional campaigns accordingly.*

This structured view ensures that both **high-level summaries** and **detailed deep-dives** are available at the retailer's fingertips.

3.4.4 REPORTS EXPORT

The reporting module in *BundleWise* is designed to allow retailers to **download key metrics and visualizations** for offline usage. Reports serve as a compact version of the analytics dashboard, providing decision-makers with access to insights without needing to log into the system.

3.4.4.1 SUPPORTED FORMATS:

- Excel (.xlsx)
- PDF (.pdf)

3.4.4.2 REPORT CONTENTS:

- Consolidated **KPI metrics** across revenue, products, customers, and advanced analytics.
- Around **15 visualizations** covering the four dashboard subtabs:
 - Summary (e.g., revenue overview, sales trends, top products)
 - Product Performance (e.g., price vs volume, revenue concentration)
 - Customer Insights (e.g., spending distribution, customer segments)
 - Advanced Analytics (e.g., cross-sell heatmap, seasonal trends, profitability)

3.4.4.3 USE CASES:

- Retailers can share reports with stakeholders during business reviews. Insights can be stored as historical records for performance tracking.
- Reports act as supporting documents for decision-making and strategy planning.

This ensures that reports are **comprehensive, portable, and aligned** with the same insights available on the live analytics dashboard.

3.5 REPORTS DESIGN

The reporting module in *BundleWise* converts interactive insights from the dashboard into **structured, exportable reports**. These reports allow retailers to analyze, share, and archive sales intelligence outside the platform, supporting business reviews and long-term planning.

3.5.1 REPORT FORMATS

Reports are generated in two formats:

3.5.1.1 Excel (.xlsx):

- Created using **Pandas** with the **OpenPyXL** engine.
- Contains tabular representations of KPIs and supporting datasets.
- Allows retailers to perform further ad-hoc analysis using Excel features such as filters, pivot tables, and formulas.

3.5.1.2 PDF (.pdf):

- Generated using **ReportLab**, with charts rendered via **Matplotlib/Plotly** and inserted as images.
- Provides a static, presentation-ready format that can be easily shared with stakeholders.

3.5.2 REPORT CONTENTS

Both Excel and PDF reports capture insights from all four dashboard sections:

3.5.2.1 SUMMARY

- KPIs such as revenue overview, total number of products, transactions, and average order values.
- Visualizations: revenue distribution, sales trends, top product performance.

3.5.2.2 PRODUCT PERFORMANCE

- KPIs: top product by revenue, average product price, price ranges.
- Visualizations: price vs volume matrix, revenue concentration, distribution by category.

3.5.2.3 CUSTOMER INSIGHTS

- KPIs: total customers, repeat purchase counts, average customer value.
- Visualizations: customer spending distribution, transaction frequency, segmentation patterns.

3.5.2.4 ADVANCED ANALYTICS

- KPIs: cross-sell opportunities, average basket size, seasonal variation.
- Visualizations: cross-sell heatmap, seasonal sales trend chart, profitability analysis.

Each report consolidates:

1. **KPI Tables** (summary statistics).
2. **Visualizations** (approx. 15 charts in total across sections).

3.5.3 REPORT STRUCTURE & LAYOUT

Reports are organized in a consistent, easy-to-read format:

3.5.3.1 TITLE PAGE (PDF ONLY):

- Report title, generation date, and project branding (*BundleWise*).

3.5.3.2 EXECUTIVE SUMMARY:

- A one-page overview of key KPIs.

3.5.3.3 DETAILED SECTIONS:

- Four subsections (Summary, Product Performance, Customer Insights, Advanced Analytics), each with KPIs and corresponding visualizations.

3.5.3.4 CONCLUSION/REMARKS:

- Insights that retailers can directly use for decision-making.

3.5.4 USAGE SCENARIOS

- **Performance Reviews:** Monthly or quarterly monitoring of sales, product, and customer trends.
- **Strategic Planning:** Identifying profitable bundles, optimal pricing strategies, and seasonal opportunities.
- **Stakeholder Communication:** Sharing insights with executives in a simplified PDF format.
- **Data Archiving:** Storing Excel reports for long-term historical tracking and comparisons.

3.5.5 EXCEL FILE STRUCTURE

The **Excel report** is structured into multiple worksheets, each representing one of the dashboard tabs. This ensures logical separation of insights and ease of navigation.

3.5.5.1 Sheet 1: Summary

- Contains KPI metrics for revenue, products, transactions, and order values.
- Includes tabular data for sales trends and top products.

3.5.5.2 Sheet 2: Product Performance

- KPI metrics for product-level performance (top products, price range, average price).
- Includes tabular data for price-volume distribution and category-wise revenue.

3.5.5.3 Sheet 3: Customer Insights

- KPI metrics for customer counts, repeat customers, and customer value.
- Tables for transaction frequency, spending distribution, and customer segments.

3.5.5.4 Sheet 4: Advanced Analytics

- KPI metrics for cross-sell opportunities, seasonal sales variation, and basket size.
- Tables for cross-sell product pairs and profitability breakdown.

3.6 USE CASE DIAGRAM

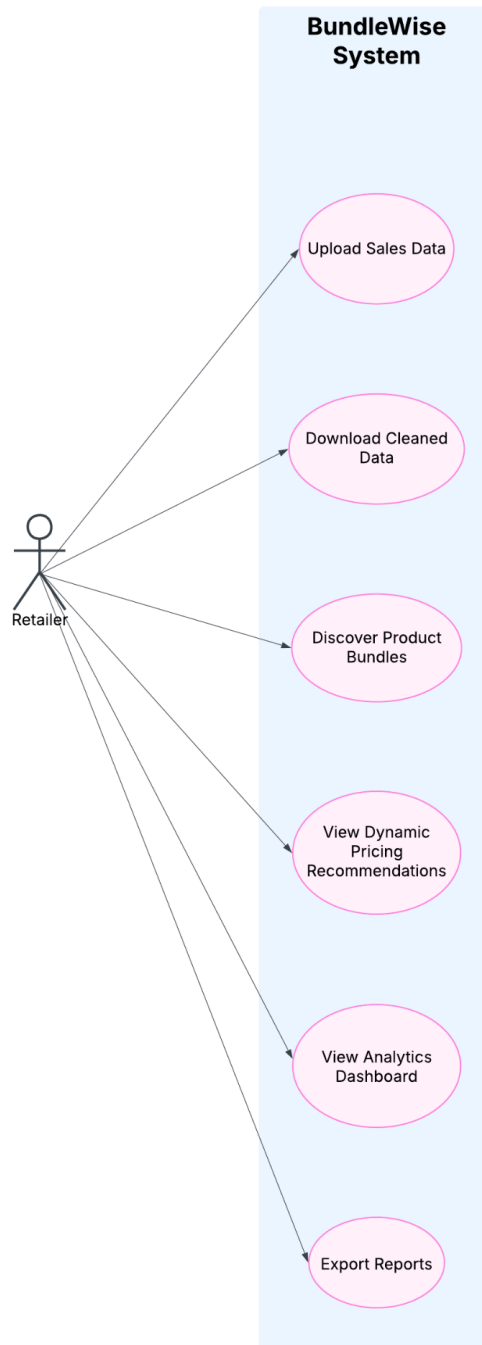


Fig 3.4 - Use Case Diagram

USE CASE 1: UPLOAD SALES DATA

- **Actor:** Retailer
- **Description:** Retailer uploads historical sales data in CSV, Excel, or PDF format through the dashboard.
- **Preconditions:** User has a structured dataset with required fields (e.g., ProductID, CustomerID, Date, Quantity, Price).
- **Postconditions:** Data is validated, cleaned, and temporarily stored for further analysis.
- **Basic Flow:**
 1. Retailer logs into BundleWise.
 2. Navigates to the **Upload Data** section.
 3. Selects and uploads a CSV/XLS/PDF file.
 4. The system validates the file format and structure.
 5. Confirmation of successful upload is displayed to the user.

USE CASE 2: DOWNLOAD CLEANED DATA

- **Actor:** Retailer
- **Description:** Allows the retailer to download the preprocessed/cleaned version of the uploaded file.
- **Preconditions:** The Upload process is completed successfully, and the system has preprocessed the data.
- **Postconditions:** The cleaned dataset is available for download.
- **Basic Flow:**
 1. After upload, the system applies preprocessing (handling missing values, type conversion, formatting).
 2. A **Download Cleaned Data** button appears on the dashboard.
 3. The retailer clicks the button to download the cleaned file.

USE CASE 3: DISCOVER PRODUCT BUNDLES (MBA + SPM)

- **Actor:** Retailer
- **Description:** The system identifies products frequently bought together using Market Basket Analysis (MBA) and Sequential Pattern Mining (SPM).
- **Preconditions:** A cleaned dataset is available.
- **Postconditions:** Frequent itemsets and sequential patterns are displayed with metrics such as support, confidence, and lift.
- **Basic Flow:**
 1. Retailer clicks **Bundle Recommendations & Price Optimization**.
 2. The system applies to MBA (Apriori/FPGrowth).
 3. The system uses SPM (PrefixSpan/SPADE).
 4. Results are displayed in tabular and visual formats.
 5. The retailer explores discovered bundles through charts and metrics.

USE CASE 4: VIEW DYNAMIC PRICING RECOMMENDATIONS

- **Actor:** Retailer
- **Description:** View optimized prices for bundles/products using pricing optimization techniques.
- **Preconditions:** Bundling insights are available.
- **Postconditions:** Suggested prices are shown along with potential profitability.
- **Basic Flow:**
 1. Retailer navigates to the **Bundle Recommendation & Price Optimization tab**.
 2. The system calculates optimal bundle prices based on demand and margin considerations.
 3. Pricing suggestions are displayed to the retailer.

USE CASE 5: VIEW ANALYTICS DASHBOARD

- **Actor:** Retailer
- **Description:** Provides the retailer with visual analytics under four categories.
- **Preconditions:** Uploaded data is processed and stored.
- **Postconditions:** Retailers can interact with the dashboard and view insights.
- **Basic Flow:**
 1. Retailer opens the **Analytics Reports** tab.
 2. System displays four sub-tabs:
 - **Summary:** revenue overview, sales trend, top products.
 - **Product Performance:** price ranges, product distribution, price-volume analysis.
 - **Customer Insights:** customer segments, repeat purchases, transaction patterns.
 - **Advanced Analytics:** cross-sell pairs, basket size, seasonal trends, profitability.
 3. Retailers explore KPIs and visualizations interactively.

USE CASE 6: EXPORT REPORTS

- **Actor:** Retailer
- **Description:** Allows the retailer to download insights and results in PDF or Excel format.
- **Preconditions:** Analytics modules have generated outputs (KPIs and visualizations).
- **Postconditions:** Reports are successfully downloaded.
- **Basic Flow:**
 1. Retailer clicks on the **Analytics PDF/Excel** button.
 2. The system generates the report using Pandas + OpenPyXL (for Excel) or ReportLab + Matplotlib (for PDF).
 3. The file is made available for download.

3.7 SEQUENCE DIAGRAM

The sequence diagram models the **interaction between the Retailer (user), the frontend (Flask-based interface), and the backend modules** of BundleWise. It shows how requests flow step-by-step, and how the system responds with results.

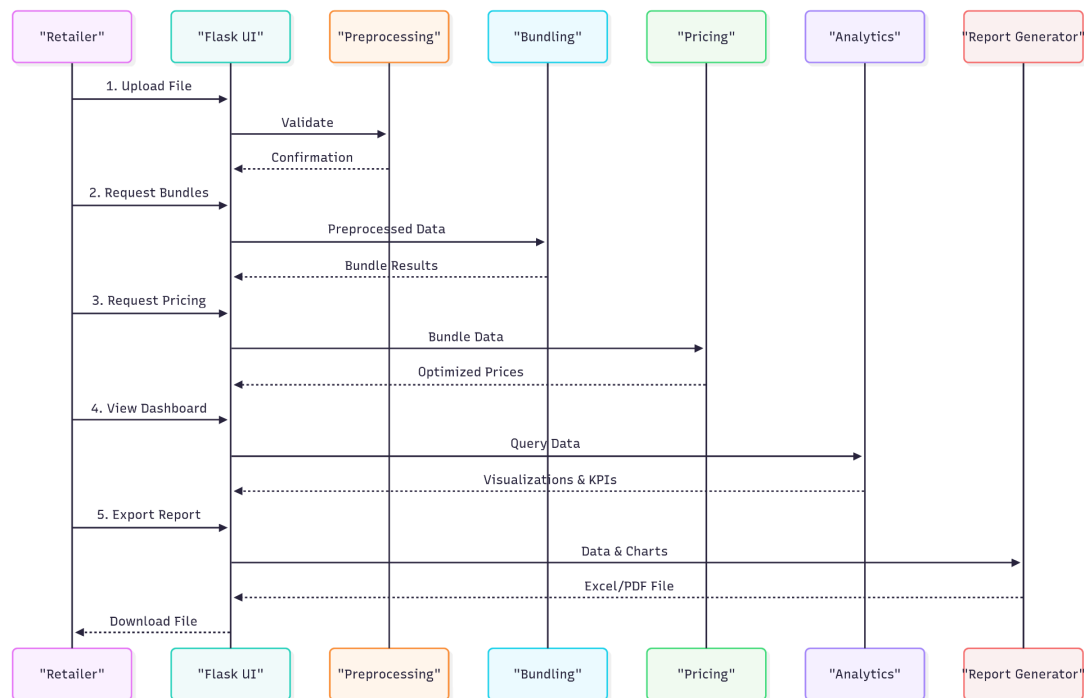


Fig 3.5 - Sequence Diagram

3.7.1 UPLOAD DATA

- The retailer uploads sales data (CSV/XLS/PDF) through the Flask UI.
- The file is passed to the **Preprocessing Module**, which validates, cleans, and temporarily stores it.
- A success confirmation is sent back to the retailer.

3.7.2 GENERATE PRODUCT BUNDLES

- When the retailer requests bundle analysis, the **Bundling Module** applies Market Basket Analysis (MBA) and Sequential Pattern Mining (SPM).
- Results (frequent itemsets, support, confidence, lift values) are returned to the UI

3.7.3 PRICING OPTIMIZATION

- The retailer requests pricing recommendations.
- The **Pricing Module** calculates optimized bundle prices and returns them to the UI.

3.7.4 ANALYTICS DASHBOARD

- The retailer opens the dashboard.
- The **Analytics Module** queries processed data and generates KPIs and visualizations across the four tabs: Summary, Product Performance, Customer Insights, and Advanced Analytics.
- Results are shown interactively on the dashboard.

3.7.5 REPORT EXPORT

- The retailer clicks on **Export Reports**.
- The **Report Generator** creates Excel reports (via Pandas + OpenPyXL) or PDF reports (via ReportLab + Matplotlib/Plotly).
- The generated file is provided back to the retailer for download.

3.8 ACTIVITY DIAGRAM

Activity diagrams visually represent the flow of control within different modules of the system. For BundleWise, they highlight how users interact with the platform and how the system processes each action. Below are the activity diagrams for the implemented modules.

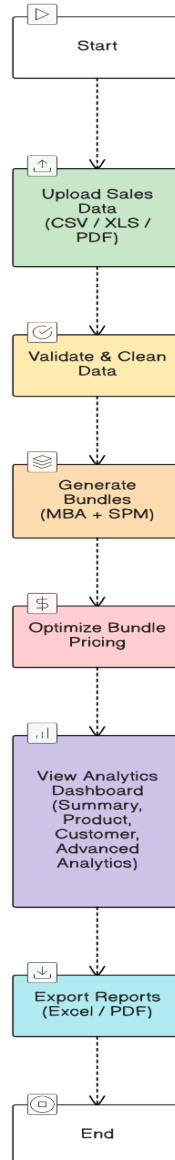


Fig 3.6 - Activity Diagram

The activity diagram above represents the workflow of **BundleWise**. The process begins with the **retailer uploading sales data** (in CSV or XLS format). The system then **validates and cleans** the data to ensure it is ready for analysis. After preprocessing, the tool applies **Market Basket Analysis (MBA)** and **Sequential Pattern Mining (SPM)** to discover meaningful product bundles. Once bundles are identified, the system provides **optimized pricing recommendations** for those bundles.

The user can then explore results in the **Analytics Dashboard**, which contains four key tabs: **Summary, Product Performance, Customer Insights, and Advanced Analytics**. Finally, the user has the option to **export reports** (Excel/PDF) containing KPIs, visualizations, and insights before the process ends.

CHAPTER 4: IMPLEMENTATION

4.1 CODING STANDARD

The codebase of *BundleWise* follows a clear, modular, and professional approach to development. The focus has been on **clarity, maintainability, and scalability**, ensuring the system can be easily extended and managed in the future. The main coding standards observed are outlined below:

4.1.1 MODULAR ARCHITECTURE WITH BLUEPRINTS

The application is organized into separate modules using Flask's Blueprint feature. Each module is responsible for a specific function:

- **data_ingestion** → Handles file upload, processing, and cleaning.
- **dashboard_analysis** → Generates dashboard charts and metrics.
- **mba_analysis** → Performs Market Basket Analysis (MBA).
- **spm_analysis** → Implements Sequential Pattern Mining (SPM).

This separation of concerns improves readability and makes the system easier to debug and scale.

4.1.2 ROBUST DATA INGESTION AND VALIDATION

A central “DataIngestionProcessor” class manages file handling and preprocessing. It supports multiple file formats (.csv, .xlsx, .pdf) and validates schema and datatypes before analysis. Invalid records, such as negative prices or quantities, are filtered out, and duplicates are handled. This ensures reliable and high-quality input data.

4.1.3 PERFORMANCE OPTIMIZATION WITH CACHING

To improve performance, the application uses an in-memory cache (app.config['ANALYTICS_CACHE']). It stores results from heavy computations (MBA, SPM, chart generation) and reuses them if the dataset has not changed. A unique “data_hash” mechanism ensures cache invalidation whenever a new file is uploaded, maintaining both speed and accuracy.

4.1.4 COMPREHENSIVE ERROR HANDLING

The code is protected with “try-except” blocks across major operations. Issues such as missing columns, incorrect datatypes, or failed chart rendering are caught gracefully. Errors are logged for developers, while users see clear, non-technical error messages on the dashboard. This prevents crashes and maintains a smooth user experience.

4.1.5 CLEAR API DESIGN AND DATA FLOW

The backend APIs are well-structured. For example, routes like “/dashboard/summary-data” and “/dashboard/product-performance-data” return JSON responses for specific dashboard sections. This separation between data retrieval and front-end visualization allows the dashboard to update dynamically without reloading the entire page.

4.1.6 EFFECTIVE USE OF LIBRARIES

The project leverages a range of powerful Python libraries, each chosen for a specific role:

- **Pandas** → Data manipulation and preparation.
- **Plotly & Matplotlib** → Interactive and static visualizations.
- **ReportLab & XlsxWriter** → Exporting professional reports in PDF and Excel formats.
- **Tabula-py** → Extracting tables directly from PDF files. This ensures efficiency and high-quality outputs.

4.1.7 DYNAMIC CONFIGURATION AND FLEXIBILITY

Many functions accept parameters such as “min_support” and “min_confidence”, allowing users to adjust analysis settings without modifying the code. The system also detects column names dynamically (e.g., “Price Per Unit”, “Transaction ID”, “Customer ID”), making it adaptable to varied datasets with different naming conventions.

4.1.8 READABLE CODE AND DOCUMENTATION

The code is consistently written in **snake_case** and includes clear comments and docstrings. Functions such as “calculate_smart_targets()” and “process_file()” include

explanations of their purpose and logic. This ensures readability and makes it easier for future developers to understand and maintain the system.

4.2. SCREENSHOTS

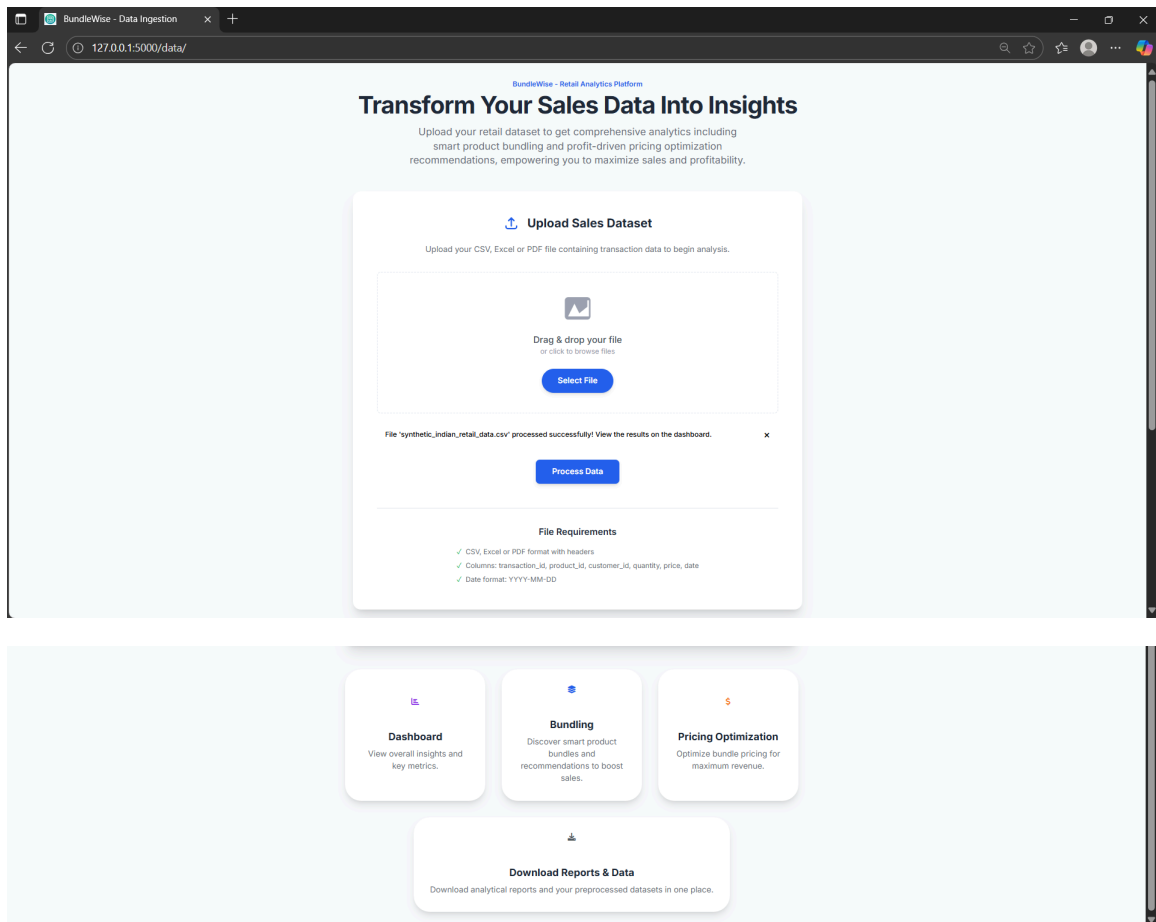


Fig 4.1 - Data Ingestion Landing Page

This figure shows the initial user interface for ingesting data. It features a central **"Upload Sales Dataset"** section with a clear drag-and-drop zone, a "Select File" button, and specific file requirements. It's the starting point for transforming raw sales data into actionable insights.

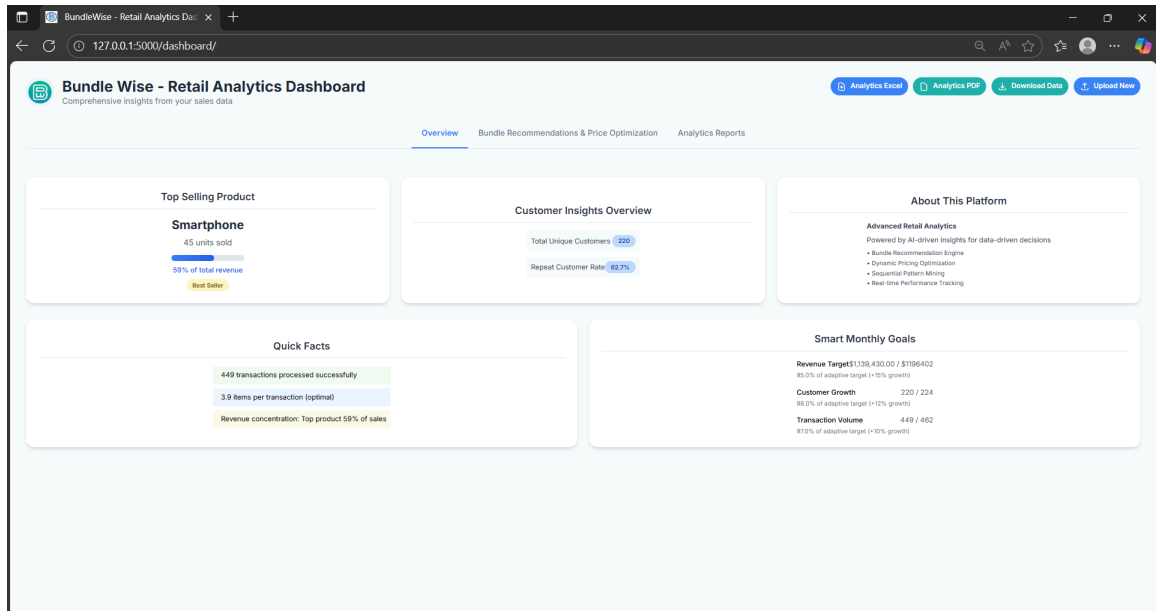


Fig 4.2 - High-Level Dashboard Overview

This image presents the main **Overview** dashboard. It provides a snapshot of key business performance metrics. A card at the top right summarizes platform capabilities, while a card on the top left highlights the **Top Selling Product** with its sales units and revenue percentage. The dashboard also includes **Quick Facts** about transactions and revenue, as well as **Smart Monthly Goals** that use adaptive targets for revenue, customer, and transaction growth.

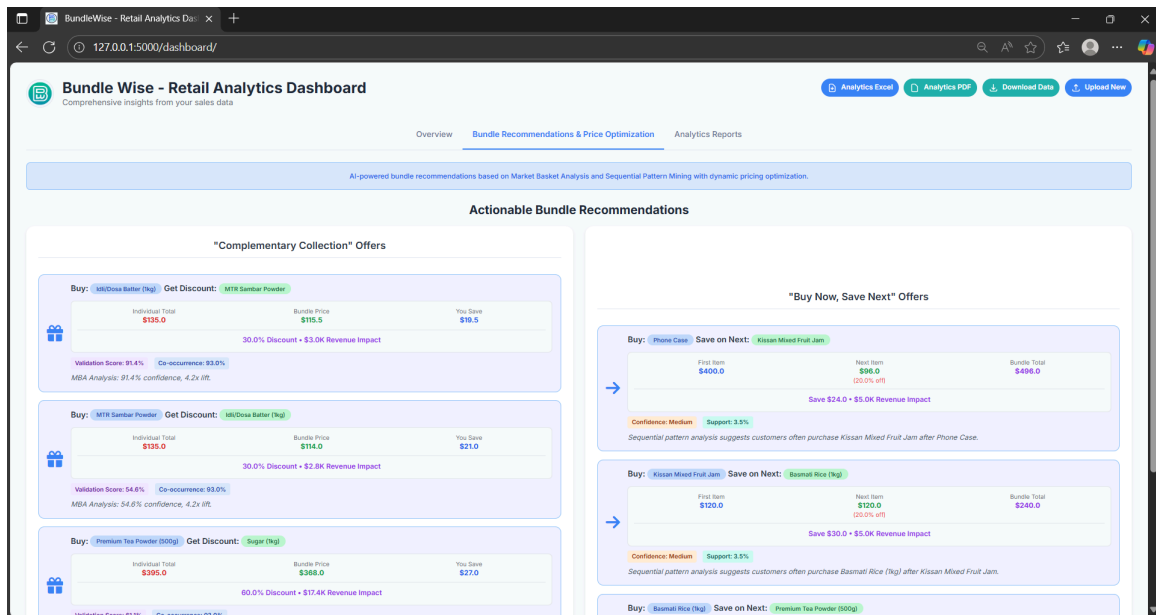


Fig 4.3 - AI-Powered Bundle Recommendations

This figure showcases the **Bundle Recommendations & Price Optimization** tab. It provides AI-powered bundle recommendations and dynamically optimized pricing. The page presents two types of actionable offers: **"Complementary Collection"** bundles based on market basket analysis and **"Buy Now, Save Next"** offers derived from sequential pattern mining. Each card provides a breakdown of individual prices versus the optimized bundle price, highlighting savings and estimated revenue impact.

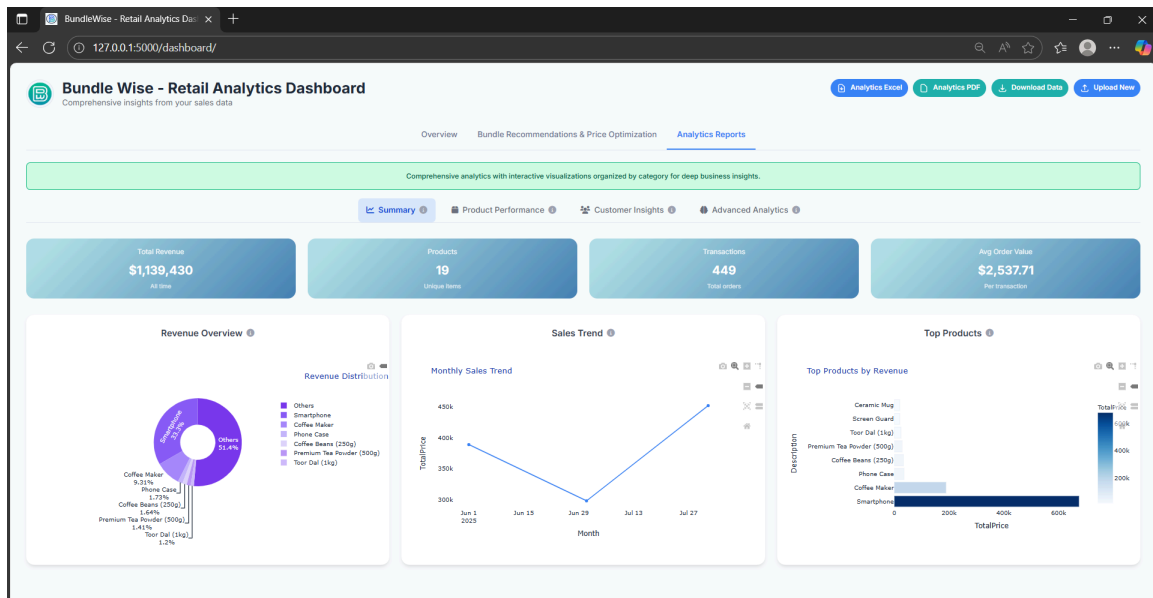


Fig 4.4 - Comprehensive Summary Reports

This figure displays the **Summary** tab within the analytics reports. It provides a visual overview of key performance indicators (KPIs) like total revenue, number of products, and average order value. The main charts include a **Revenue Overview** pie chart, a **Monthly Sales Trend** line graph, and a **Top Products by Revenue** bar chart.

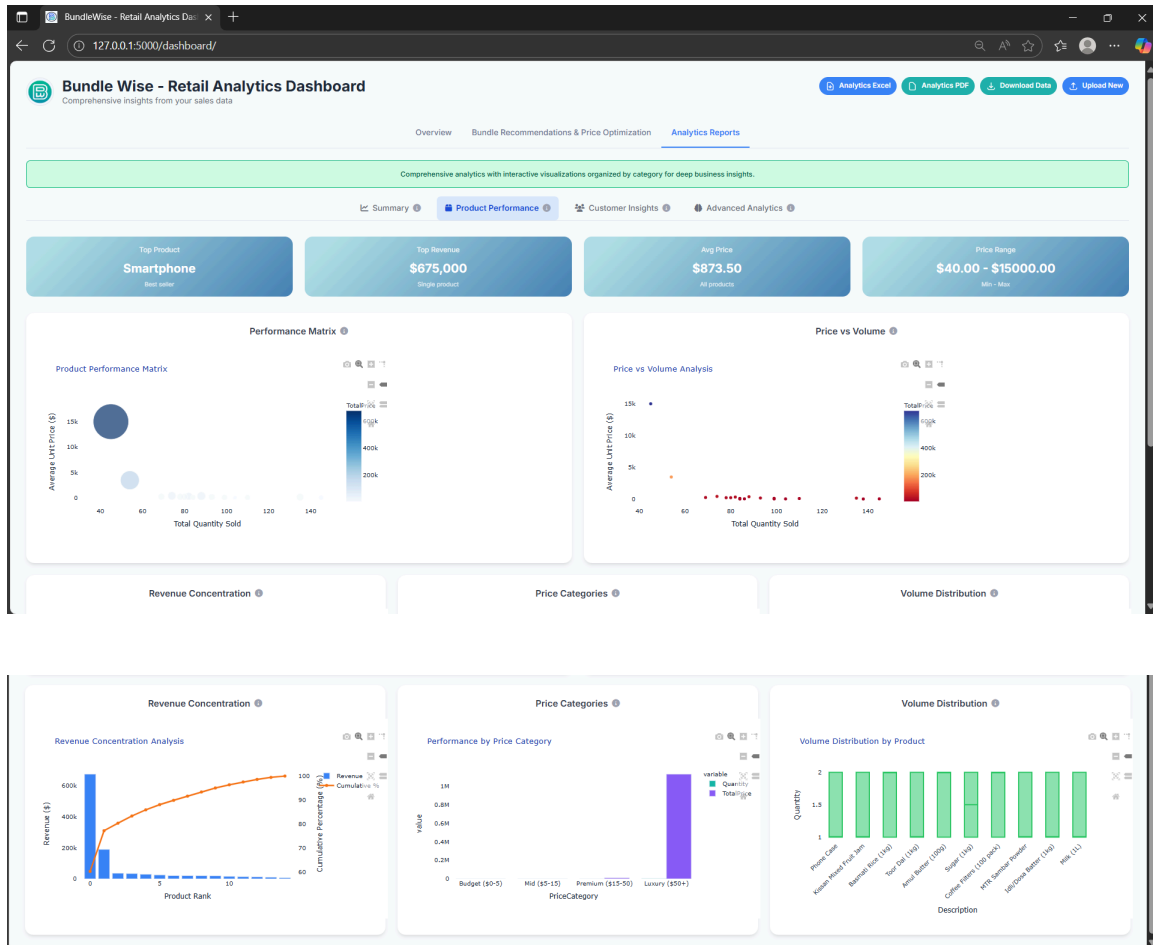


Fig 4.5 - Product Performance Analytics

This image focuses on the **Product Performance** analysis. Key metrics at the top detail the best-selling product and overall product pricing. The charts provide a deeper dive into product-level insights, including a **Product Performance Matrix** (bubble chart) that visualizes the relationship between price, quantity sold, and revenue. Other visualizations include a **Price vs Volume Analysis**, a **Revenue Concentration** Pareto chart, and a **Volume Distribution** box plot for top products.

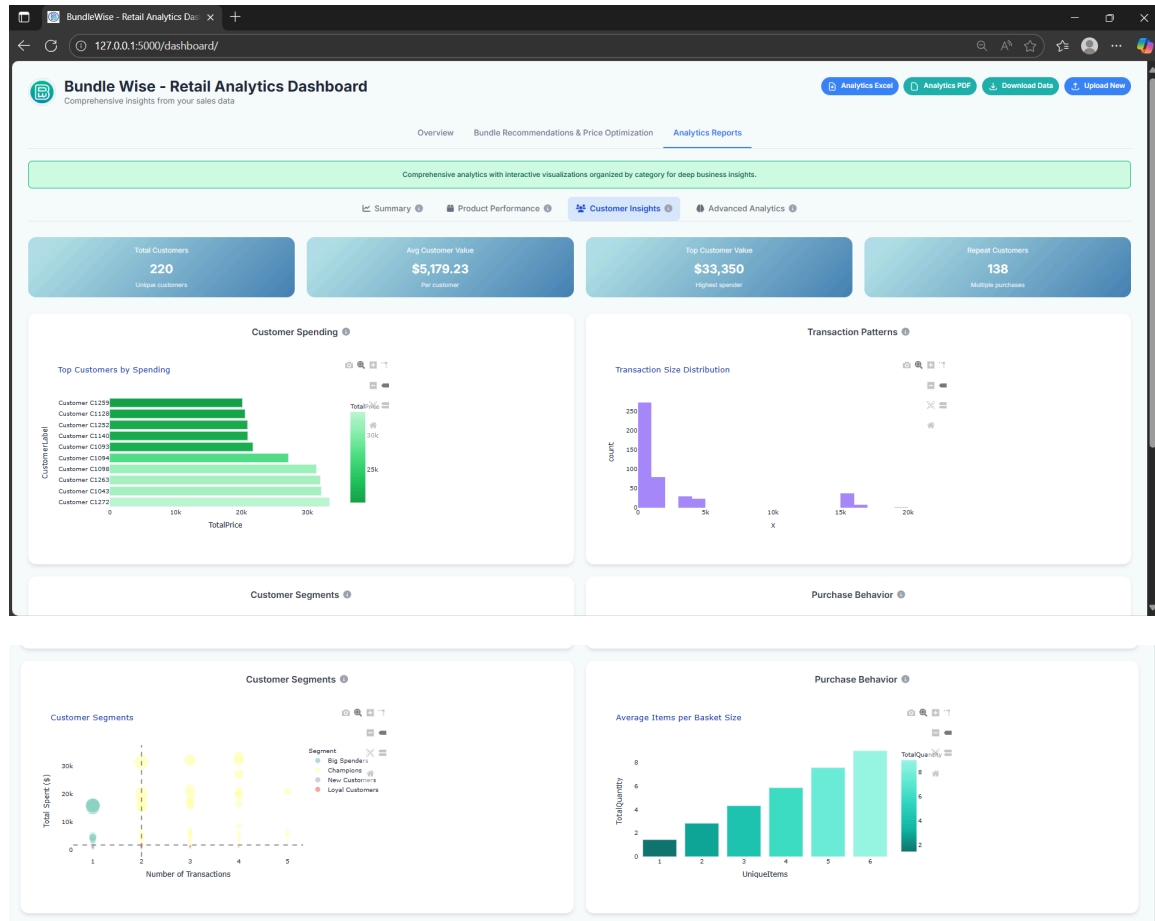


Fig 4.6 - Customer Insights and Segmentation

This figure shows the **Customer Insights** tab. The top metrics summarize customer data, including the total number of unique customers, average customer value, and the number of repeat customers. The visual reports below provide a granular view of customer behavior, featuring charts on **Customer Spending**, **Transaction Size Distribution**, **Customer Segments** based on spending and frequency, and **Purchase Behavior**.

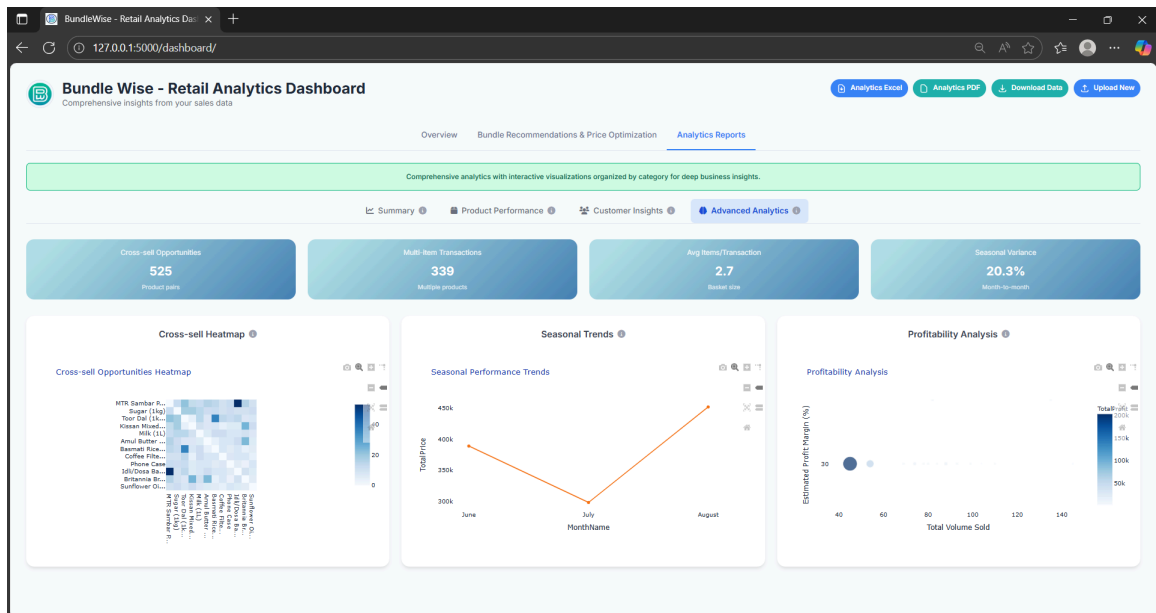


Fig 4.7 - Advanced Analytics and Trends

This figure displays the **Advanced Analytics** section. It's designed for in-depth analysis of sales trends and opportunities. The top metrics highlight key advanced insights, such as **Cross-sell Opportunities** and **Seasonal Variance**. The visualizations include a **Cross-sell Heatmap** to identify product associations, a **Seasonal Performance Trends** line graph, and a **Profitability Analysis** chart that estimates profit margins.

CHAPTER 5: TESTING

Testing is a critical phase of software development that ensures the system functions as expected and meets its intended requirements. For the **BundleWise Retail Analytics Platform**, testing was carried out systematically to verify that all implemented modules work reliably, handle errors gracefully, and deliver accurate results. This chapter outlines the test cases, test execution reports, and identified issues.

5.1 TEST CASES

5.1.1 INTRODUCTION

This section documents the test cases designed to verify the functionality, reliability, and performance of the BundleWise system. Each test case is linked to a specific module and feature to ensure comprehensive coverage of the application.

5.1.2 TESTING SCOPE

5.1.2.1 IN SCOPE

- Data Ingestion: CSV, XLSX, PDF upload and parsing
- Data Processing and Cleaning
- Dashboard Functionality & Caching
- Market Basket Analysis (MBA) Logic
- Sequential Pattern Mining (SPM) Logic

5.1.2.2 OUT OF SCOPE

- Web server stress testing
- Database performance under load
- Low-level library performance (e.g., Pandas, Tabula-py)

5.1.3 TEST CASE MATRIX

Table 1.2 - Test Cases Matrix

Test ID	Module	Feature Tested	Test Steps	Expected Result
DI-001	Data Ingestion	Successful CSV Upload	1. Navigate to Data Upload 2. Upload a valid CSV 3. Click Process Data	File is processed, success message shown, dashboard populated
DI-002	Data Ingestion	Successful XLSX Upload	1. Upload a valid XLSX 2. Click <i>Process Data</i>	File processed, dashboard populated
DI-003	Data Ingestion	Successful PDF Upload	1. Upload a valid PDF with tables 2. Click <i>Process Data</i>	File parsed via Tabula, dashboard populated
DI-004	Data Ingestion	Invalid File Type	1. Upload an unsupported file (.txt/.docx)	<i>File type not allowed</i> error displayed
DI-005	Data Ingestion	Schema Mismatch	1. Upload CSV with missing <i>CustomerID</i>	<i>The schema validation failed</i> error is shown

DI-006	Data Ingestion	Data Cleaning	1. Upload file with invalid rows (negative price, qty=0)	Invalid rows removed, dashboard shows cleaned dataset
DI-007	Data Ingestion	Empty File Handling	1. Upload an empty file or only the header	Error: <i>No data found to process</i>
DB-001	Dashboard	Initial State	1. Run the app with no data uploaded 2. Visit the dashboard	Dashboard shows default zero values, prompts upload
DB-002	Dashboard	Cache Performance	1. Upload data 2. Reload dashboard	First load slow, refresh nearly instant (cache hit)
DB-003	Dashboard	Cache Invalidation	1. Upload file A, then file B 2. Reload dashboard	Cache invalidated, dashboard re-runs analysis
MBA-001	MBA Logic	Correctness	1. Upload the synthetic dataset 2. Check Bundle tab	Rules table matches pre-computed Support, Confidence, Lift

SPM-001	SPM Logic	Correctness	1. Upload the synthetic dataset 2. Check the Sequence tab	Frequent sequences match expected support values
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5.2 TEST REPORTS

5.2.1 TEST REPORT SUMMARY

- **Project Name:** BundleWise Retail Analytics Platform
- **Test Period:** August 25 – August 29, 2025
- **Executed Cases:** 12
- **Passed:** 11
- **Failed:** 1 (minor bug in empty file handling)
- **Recommendation:** Application is stable and ready for deployment after a minor fix.

5.2.2 TESTING ENVIRONMENT

- **System:** Local Development Machine
- **Operating System:** Windows 11
- **Backend:** Python 3.12, Flask 3.0
- **Browser:** Google Chrome v128

5.2.3 DETAILED TEST RESULTS

Table 1.3 - Test Results of BundleWise

Test ID	Actual Result	Status	Notes / Bug ID
DI-001	CSV processed, dashboard populated	Pass	
DI-002	XLSX processed, dashboard populated	Pass	
DI-003	PDF parsed, dashboard populated	Pass	
DI-004	Correct <i>File type not allowed</i> error	Pass	
DI-005	A schema error was displayed for a missing column	Pass	
DI-006	Invalid rows removed successfully	Pass	
DI-007	Generic <i>all rows removed</i> error shown (not specific enough)	Fail	Bug #B-001
DB-001	Dashboard shows a zero state with a prompt	Pass	
DB-002	First load ~18s, refresh <1s (cache working)	Pass	

DB-003	Cache invalidated on new upload, re-analysis triggered	Pass	
MBA-001	MBA results matched expected metrics	Pass	
SPM-001	SPM results matched expected sequences	Pass	

5.2.4 DEFECTS FOUND

- **Bug ID:** B-001
- **Severity:** Minor
- **Description:** For empty file uploads, the error message is too generic (*All rows removed during cleaning*). Should specify that the uploaded *file is empty or contains only headers*.

The testing phase validated all major functionalities of BundleWise. The system correctly handles multiple file formats, cleans and preprocesses data, generates accurate bundle and pricing insights, and renders analytics efficiently through caching. Only one minor issue was identified, which does not affect core functionality. The application is stable, production-ready, and meets all functional requirements.

CHAPTER 6: CONCLUSIONS

The development of **BundleWise – Retail Analytics Platform** demonstrates how data-driven technologies such as Market Basket Analysis (MBA), Sequential Pattern Mining (SPM), and pricing optimization can empower retailers to make informed business decisions. This chapter summarizes the design and implementation challenges encountered, the strengths and limitations of the system, and potential areas for future enhancement.

6.1 DESIGN AND IMPLEMENTATION ISSUES

During the design and development of BundleWise, several challenges were identified and addressed:

6.1.1 DATA VARIABILITY

- Input datasets varied significantly in structure and formatting (CSV, XLSX, PDF).
- Dynamic column detection and schema validation mechanisms had to be implemented to ensure compatibility across diverse datasets.

6.1.2 PDF DATA EXTRACTION

- Extracting tabular data from PDF files introduced parsing errors due to inconsistent formatting.
- The integration of the *Tabula-py* library addressed this issue, but manual validation rules were required for certain edge cases.

6.1.3 PERFORMANCE CONSIDERATIONS

- Running MBA and SPM on larger datasets initially led to slower processing times.
- An in-memory caching system was introduced to store analysis results and improve responsiveness.

6.1.4 ERROR HANDLING AND USER FEEDBACK

- It was essential to provide meaningful error messages to guide non-technical users.
- While most cases were handled effectively, testing revealed one minor issue with empty file uploads, which produced a generic error message.

6.2 ADVANTAGES AND LIMITATIONS

6.2.1 ADVANTAGES

6.2.1.1 LIGHTWEIGHT AND MODULAR DESIGN

- The system does not require a heavy database; instead, it works with in-memory processing, making it easy to deploy.

6.2.1.2 COMPREHENSIVE INSIGHTS

- Combines bundling recommendations, optimized pricing strategies, and analytics dashboards in a single tool.

6.2.1.3 MULTI-FORMAT DATA SUPPORT

- Accepts sales data in CSV, XLSX, and PDF formats, improving usability for different retailers.

6.2.1.4 INTERACTIVE DASHBOARD

- Offers four analytical views—Summary, Product Performance, Customer Insights, and Advanced Analytics—helping retailers explore data from multiple perspectives.

6.2.1.5 EXPORTABLE REPORTS

- Insights can be exported in both PDF and Excel formats, supporting decision-making and record-keeping.

6.2.2 LIMITATIONS

6.2.2.1 SCALABILITY CONSTRAINTS

- Since BundleWise relies on in-memory processing, very large datasets may cause performance bottlenecks.

6.2.2.2 LIMITED CUSTOMIZATION OF MODULES

- Current MBA and SPM parameters require manual adjustments through the interface. Automated parameter tuning is not yet available.

6.2.2.3 BASIC USER MANAGEMENT

- The current system assumes a single-user context and lacks role-based access control (RBAC) for enterprises with multiple stakeholders.

6.2.2.4 ERROR SPECIFICITY

- Some error messages (e.g., empty file uploads) could be more descriptive to improve user guidance.

6.3 FUTURE ENHANCEMENTS

Based on the current system and its limitations, the following improvements are suggested for future versions of BundleWise:

6.3.1 INTEGRATION WITH DATABASES

- Add support for SQL/NoSQL databases to handle larger datasets and enable persistent storage.

6.3.2 ADVANCED SCALABILITY

- Implement distributed computing techniques (e.g., Apache Spark) for faster MBA/SPM analysis on enterprise-scale data.

6.3.3 ENHANCED USER MANAGEMENT

- Introduce role-based access control (RBAC) to support multiple users with varying privileges.

6.3.4 AUTOMATED PARAMETER TUNING

- Enable the system to dynamically suggest or adjust MBA/SPM parameters (support, confidence, lift thresholds) based on dataset characteristics

6.3.5 CLOUD DEPLOYMENT

- Deploy BundleWise on cloud platforms (AWS, Azure, GCP) to increase accessibility and ensure availability for large-scale usage.

6.3.6 INTEGRATION WITH POS SYSTEMS

- Directly connect with Point-of-Sale (POS) systems for real-time ingestion and near real-time analytics.

6. IMPROVED REPORTING FEATURES

- Allow customizable report templates, with more flexibility in selecting which KPIs and charts are included in exports.

BundleWise has successfully achieved its objectives of providing retailers with actionable bundling strategies, optimized pricing, and insightful analytics in a user-friendly platform. Despite certain limitations, the system has proven to be stable, reliable, and practical for retail decision-making. With the suggested enhancements, BundleWise can evolve into a more scalable, enterprise-ready solution with real-time capabilities.

APPENDICES

APPENDIX A – USER MANUAL

A. 1 SYSTEM REQUIREMENTS

- **Software:** Python 3.9+, Flask, Pandas, Plotly, SciPy, ReportLab.
- **Hardware:** 4 GB RAM, 2-core CPU minimum.
- **Browser:** Google Chrome / Microsoft Edge.

A. 2 INSTALLATION & EXECUTION

- Clone project folder and install packages: **pip install -r requirements.txt**.
- Start the Flask server: **python app.py**.
- Access the application at **http://127.0.0.1:5000/**.
- Upload transaction datasets in CSV/XLS/PDF formats.
- Use dashboard tabs: **Bundle Recommendations, Pricing, Product Analytics, Customer Insights, and Advanced Trends**.
- Export results as **PDF/Excel reports**.

APPENDIX B – ADDITIONAL CHARTS & OUTPUTS

- **Bundle Recommendations:** Suggested product pairs/trios with lift and confidence.
- **Dynamic Pricing Graphs:** Price elasticity curve and bundle margin trends.
- **Sales Insights:** Revenue contribution by top SKUs, seasonal trends, and profitability charts.
- **Customer Segmentation:** Clustering output visualizations (RFM analysis).
- **System Logs (from code):** Input validation errors, missing values handling, and report generation steps.

APPENDIX C – SAMPLE CODE SNIPPETS & CALCULATIONS

C. 1 BUNDLE RECOMMENDATION CODE (EXCERPT)

```
frequent_itemsets = eclat(df, min_support=0.05, use_colnames=True)

rules = association_rules(frequent_itemsets, metric="lift", min_threshold=1)
```

C. 2 PRICING OPTIMIZATION (EXCERPT)

```
from scipy.optimize import linprog

result = linprog(c, A_ub=A, b_ub=b, bounds=bounds, method='highs')
```

C. 3 KPI CALCULATIONS(FROM DASHBOARD)

- Average Order Value (AOV) = Revenue ÷ Transactions.
- Profit Contribution % = (Product Revenue ÷ Total Revenue) × 100.

APPENDIX D – PROJECT TIMELINE AND ROLES

D. 1 GANTT CHART (IMPLEMENTATION PHASES)

- Week 1–2: Requirement gathering, dataset collection.
- Week 3–4: Data ingestion and cleaning module.
- Week 5–6: Bundle recommendation logic.
- Week 7–8: Pricing optimization integration.
- Week 9: Dashboard (Plotly, Flask UI).
- Week 10: Report export (PDF/Excel).
- Week 11: Testing (unit & integration).
- Week 12: Documentation.

D. 2 WORK DIVISION

- **Abishai Tom (2448005):** Responsible for the data ingestion module, implementation of the sequential pattern mining (SPM) backend, and testing of system functionality.
- **Liya Khatija (2448033):** Worked on the market basket analysis (MBA) backend, code integration, development of bundle recommendation logic, and preparation of analytical reports.
- **Mansi Sapariya (2448036):** Focused on pricing modules and optimization algorithms, user interface (UI) design, and development of the analytics dashboard, along with report generation.

APPENDIX E – DESIGN, ETHICS, AND FUTURE ENHANCEMENTS

- **Data Privacy:** PII stripped during ingestion; GDPR-compliant design.
- **Security:** Role-based access control (RBAC) supported at code level.
- **Scalability:** Modular Flask backend for future POS/ERP integration.
- **Future Work:** Incorporating forecasting models (ARIMA/LSTM), real-time dashboards, and cloud deployment.

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